Latent State-Space Smooth Transition Models for Realized Volatility Forecasting

Yen Wee Lim, University of Malaya, Kuala Lumpur, Malaysia

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Abstract

We propose Deep-LSTR, integrating Variational Recurrent Neural Networks with Smooth Transition HAR models using inferred latent states as transition variables. Evaluating 21 assets across equity, fixed income, currency, and commodity markets (2015–2025), we demonstrate that latent regime identification yields consistent improvements over observable-transition specifications, winning on 60–80% of individual assets and achieving Model Confidence Set inclusion rates of 89–100% across forecast horizons. The latent transition variable exhibits a regime smoothing mechanism, reaching extreme values on only 19% of days compared to 34% for observable proxies, consistent with fewer false positive regime switches. We apply the Fissler-Ziegel joint elicitability framework for Expected Shortfall evaluation, achieving 1.5% lower FZ0 loss than the HAR benchmark while maintaining proper VaR calibration. Robustness checks confirm insensitivity to latent dimensionality, transition function specification, subsample period, and volatility estimator choice.

1. Introduction

Realized volatility forecasting underpins quantitative risk management, derivative pricing, and portfolio allocation. Accurate volatility predictions are essential for computing Value-at-Risk and Expected Shortfall under Basel III/IV regulatory frameworks, pricing volatility derivatives, and constructing mean-variance optimal portfolios. The Heterogeneous Autoregressive (HAR) model of Corsi (2009) serves as the dominant benchmark, elegantly approximating long-memory dynamics through daily, weekly, and monthly volatility cascades while

requiring only four parameters estimable via ordinary least squares. The HAR model's combination of parsimony, interpretability, and competitive forecasting accuracy has established it as the standard against which new volatility models are measured. However, the linear HAR structure assumes constant coefficients across market conditions, ignoring the well-documented regime-switching behavior observed during alternating periods of market tranquility and crisis.

Smooth Transition Autoregressive (STAR) models address this limitation by allowing coefficients to vary continuously between regimes according to a transition function (Teräsvirta 1994; Granger and Teräsvirta 1993). Applications to volatility forecasting have combined the STR framework with HAR specifications (McAleer and Medeiros 2008), permitting regime-dependent persistence parameters that capture the differential dynamics of calm versus turbulent markets. During high-volatility regimes, the daily component typically dominates as markets exhibit heightened short-term reactivity, while low-volatility regimes display stronger monthly persistence reflecting gradual mean-reversion. Yet existing STR-HAR implementations rely exclusively on observable transition variables—typically lagged realized volatility or past returns.

This observable-transition assumption is problematic. Observable volatility measures are inherently noisy—transient spikes from earnings announcements, flash crashes, and liquidity events can trigger false positive regime switches that do not persist. The 2010 Flash Crash and March 2020 COVID-19 liquidity crisis demonstrate that observable volatility exhibits extreme short-term movements that may not reflect true regime shifts. An observable-STR model is therefore susceptible to overreaction, classifying brief volatility spikes as regime

changes and generating unstable coefficient dynamics. This limitation motivates the search for latent transition variables that can filter noise from true regime information.

This paper proposes the Deep Latent State-Space Smooth Transition Autoregressive (Deep-LSTR) framework to resolve this limitation. We employ a Variational Recurrent Neural Network (Chung et al. 2015) to infer a continuous latent state $z_t \in \mathbb{R}^d$ from observable market signals including realized volatility, returns, and trading volume. This latent state is linearly projected to form the scalar transition variable $s_t = \alpha^\top z_t$ governing a smooth transition HAR specification. The resulting architecture preserves the interpretability of regime-dependent HAR coefficients while leveraging deep probabilistic inference to construct a smoothed transition variable that filters transient volatility noise. Crucially, the STR-HAR coefficients remain directly interpretable as regime-specific volatility persistence parameters, distinguishing our approach from pure black-box forecasting methods.

We contribute to the literature in three dimensions. First, to our knowledge, we introduce the first integration of deep variational state-space models with the interpretable STR-HAR econometric framework, demonstrating that the VRNN generalizes classical state-space econometrics while preserving coefficient interpretability. The latent state provides a data-driven transition variable without requiring the researcher to specify observable proxies ex ante. Second, we provide extensive multi-asset, multi-horizon evidence that latent regime identification dominates observable proxies through a regime smoothing mechanism that reduces false positive regime switches. Our evaluation spans 21 liquid assets across equity, fixed income, currency, and commodity markets, addressing concerns about single-asset overfitting that pervade the volatility forecasting literature. Third, we apply the joint elicitability

framework of Fissler and Ziegel (2016) to rigorously compare tail risk forecasting performance, addressing the methodological deficiency whereby most deep-learning volatility studies evaluate Expected Shortfall using improper scoring rules. To our knowledge, this represents the first valid ES comparison for deep learning volatility models.

The remainder of this paper proceeds as follows. Section 2 reviews the literature on realized volatility forecasting, regime-switching models, and deep probabilistic approaches. Section 3 develops the Deep-LSTR methodology. Section 4 describes the data and benchmark models. Section 5 presents empirical results. Section 6 discusses robustness checks. Section 7 concludes.

2. Literature Review

2.1. Realized Volatility and the HAR Model

The development of high-frequency financial econometrics enabled direct observation of a quantity traditionally treated as latent. Andersen and Bollerslev (1998) demonstrated that realized variance, computed as the sum of squared intraday returns, provides a consistent estimator of integrated variance under standard diffusion assumptions. This paradigm shift from latent-variable GARCH models to observable, model-free volatility measurement was formalized by Andersen et al. (2003), who established three empirical regularities: realized volatility exhibits long memory with slowly decaying autocorrelations, its logarithm is approximately Gaussian, and large shocks occur more frequently than Gaussian predictions imply. The importance of accurate volatility measurement extends beyond forecasting to

option pricing, portfolio allocation, and risk management (Poon and Granger 2003).

Corsi (2009) introduced the HAR-RV model, which approximates long-memory dynamics through a simple additive cascade:

$$RV_{t+H} = \beta_0 + \beta_d RV_t^{(d)} + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \epsilon_{t+H}$$

where $RV^{(d)}, RV^{(w)}, RV^{(m)}$ represent daily, weekly, and monthly realized volatility components. The theoretical foundation derives from the Heterogeneous Market Hypothesis (Müller et al. 1997), which posits that market participants operating on different time horizons generate distinct volatility components. The HAR model's parsimony and interpretability have made it the dominant benchmark in realized volatility forecasting.

The HAR framework has spawned numerous extensions addressing specific empirical phenomena. Andersen et al. (2007) introduced HAR-J separating continuous variation from jumps, recognizing that discontinuous price movements carry distinct information content. Corsi et al. (2010) extended this in HAR-TCJ with threshold-based jump detection. Bollerslev et al. (2016) augmented HAR with realized quarticity in HAR-Q, exploiting measurement error heteroskedasticity. Patton and Sheppard (2015) incorporated leverage effects through LHAR, decomposing returns into positive and negative components. Bekaert and Hoerova (2014) examined the relationship between realized and implied volatility within HAR frameworks. Despite these advances, none allows for the smooth, continuous coefficient variation that characterizes transitions between market regimes.

Recent work has challenged the traditional long-memory interpretation of volatility persis-

tence. Gatheral et al. (2018) demonstrated that realized volatility exhibits "rough" behavior characterized by a Hurst exponent $H \approx 0.1$, far below the $H = 0.5$ threshold that would indicate standard Brownian motion. This roughness—evident in the power-law decay of volatility increment autocorrelations—suggests that volatility is locally more irregular than previously assumed while maintaining long-range dependence at larger scales. The rough volatility framework, formalized through fractional Brownian motion with $H < 0.5$, has profound implications for option pricing and volatility forecasting. Comte and Renault (1998) provided theoretical foundations for long-memory stochastic volatility in continuous time, establishing conditions under which fractional integration arises from volatility of volatility dynamics. Bennedsen et al. (2022) further developed this literature by proposing methods to decouple short-term roughness from long-term persistence using Brownian semistationary processes, demonstrating that volatility exhibits both properties simultaneously at different time scales. These findings motivate multi-scale approaches to volatility modeling and support the HAR framework's decomposition of volatility into components operating at different frequencies.

2.2. Regime-Switching in Volatility

Financial volatility exhibits well-documented nonlinearities that linear specifications cannot capture. The seminal work of Engle (1982) established the Autoregressive Conditional Heteroskedasticity (ARCH) framework, demonstrating that volatility clustering—periods of high volatility followed by high volatility and low by low—can be captured through models where conditional variance depends on past squared residuals. This foundational contribution, rec-

ognized with the 2003 Nobel Prize in Economics, spawned the extensive GARCH literature (Bollerslev 1986) and established the paradigm of time-varying conditional variance that underlies all modern volatility modeling.

The leverage effect (Black 1976; Christie 1982) describes asymmetric responses whereby negative returns increase future volatility more than positive returns. Volatility feedback effects create self-exciting dynamics where high volatility generates uncertainty, affecting returns and further amplifying volatility (French et al. 1987). Most consequentially, volatility exhibits regime dependence with fundamentally different behavior during calm versus turbulent periods.

Hamilton (1989) introduced regime-switching models where an unobserved Markov chain governs parameter changes. Applications to volatility produced Markov-Switching GARCH (MS-GARCH) models (Hamilton and Susmel 1994; Gray 1996; Haas et al. 2004). While MS-GARCH captures distinct regimes, it imposes abrupt transitions and requires probabilistic regime inference. Klaassen (2002) addressed the path-dependence problem in MS-GARCH through expected regime-conditional variances, though computational complexity remains substantial.

Teräsvirta (1994) developed the Smooth Transition Autoregressive (STAR) framework allowing continuous, gradual transitions:

$$y_t = (1 - G(s_t))\Phi_1(y_{t-r}) + G(s_t)\Phi_2(y_{t-r}) + \epsilon_t$$

where $G(\cdot)$ is a bounded transition function and s_t is the transition variable. The STAR

framework nests both linear models (as $\gamma \to 0$) and threshold autoregressions (as $\gamma \to \infty$). Granger and Teräsvirta (1993) provided comprehensive guidance on specification, estimation, and evaluation of nonlinear time series models within this framework.

McAleer and Medeiros (2008) extended the HAR framework with smooth transition dynamics, proposing the Multiple-Regime Smooth Transition HAR (HARST) model. Applied to 16 DJIA stocks using high-frequency TAQ data, they demonstrated that logistic transition functions driven by lagged returns capture both long memory (through the HAR structure) and sign/size asymmetries (through regime-dependent coefficients). Their analysis revealed that regime transitions typically occur at large negative returns—around -3% to -6%—rather than at zero, with two regimes sufficient for most assets. Importantly, they found that combining HAR and HARST forecasts outperforms either model alone, suggesting complementary information content. However, this approach assumes regime transitions are identifiable from observable past data—an assumption violated when regime shifts are driven by unobserved factors including liquidity conditions, institutional positioning, and investor sentiment. The present paper addresses this limitation by inferring latent transition variables through deep state-space models.

2.3. Deep Probabilistic Models

The Variational Autoencoder framework (Kingma and Welling 2014) enabled principled deep generative modeling with probabilistic latent representations, tractable training via Evidence Lower Bound maximization, and uncertainty quantification through posterior inference. These advances opened new possibilities for financial applications requiring both

flexibility and interpretable uncertainty estimates.

Sequential extensions established deep probabilistic state-space models as a mature methodology. The Variational Recurrent Neural Network (Chung et al. 2015) incorporates latent variables at each time step, maintaining separation between latent dynamics and observations that mirrors classical state-space econometrics (Durbin and Koopman 2012) while leveraging neural parameterization. The Deep Kalman Filter (Krishnan et al. 2017) provides an alternative architecture more explicitly parameterizing state-space dynamics. Rangapuram et al. (2018) demonstrated the effectiveness of deep state-space models for probabilistic time series forecasting across diverse domains.

Machine learning approaches to financial prediction have achieved notable empirical success. Gu et al. (2020) provided comprehensive evidence that machine learning methods substantially improve cross-sectional return prediction relative to traditional factor models. Recent advances include DeepVol (Moreno-Pino and Zohren 2024), which employs dilated causal convolutions for volatility forecasting from high-frequency data, achieving state-of-the-art performance but remaining a pure prediction framework without explicit regime structure. Hu et al. (2025) proposed HAR-type models using graph neural networks to capture cross-asset dependencies, exploiting information spillovers across markets. Zhang et al. (2025) extend HAR models with graph neural networks to capture nonlinear volatility spillovers across assets, demonstrating that nonlinear cross-asset dependencies—particularly when models are trained with quasi-likelihood loss—substantially improve short-horizon forecasts.

Despite substantial advances, no prior work integrates deep state-space models with interpretable econometric frameworks while using inferred latent states as transition variables.

This methodological gap motivates the present paper's contribution. Furthermore, the deep learning volatility literature has uniformly failed to apply valid elicitability-based evaluation to tail risk forecasts, limiting the reliability of comparative claims.

2.4. Transformer Architectures for Time Series

The Transformer architecture (Vaswani et al. 2017), originally developed for natural language processing, has been adapted extensively for time series forecasting since 2021. These attention-based models offer an alternative paradigm to recurrent architectures, with several innovations addressing the specific challenges of temporal prediction.

Zhou et al. (2021) introduced the Informer architecture, which addresses the quadratic complexity of standard self-attention through ProbSparse attention, reducing computational cost from $O(L^2)$ to $O(L \log L)$ for sequence length L. Designed for Long Sequence Time Series Forecasting (LSTF), Informer demonstrated substantial improvements on electricity, weather, and traffic datasets. However, the architecture lacks explicit latent state modeling, treating temporal dynamics implicitly through attention patterns rather than through structured state representations.

Lim et al. (2021) developed the Temporal Fusion Transformer (TFT), which combines LSTM layers for local processing with interpretable multi-head attention for long-range dependencies. TFT's architecture includes variable selection networks that provide instance-wise feature importance, offering interpretability advantages over black-box alternatives. Notably, their evaluation on the Oxford-Man Institute realized volatility dataset—the same data source used in the present study—showed modest improvements of 7-8% over baselines,

substantially lower than the 20%+ gains observed on other domains. The authors found that attention weights on volatility data exhibited no persistent temporal patterns, instead acting as uniform smoothing filters, suggesting that standard Transformer architectures may not optimally capture volatility dynamics.

More recently, Nie et al. (2023) proposed PatchTST, which applies the patching strategy from vision Transformers to time series. By treating contiguous subsequences as tokens rather than individual time points, PatchTST achieves state-of-the-art performance while maintaining computational efficiency. The channel-independence design, where each variable is processed separately, reduces model complexity but sacrifices cross-variable information that may be valuable for volatility forecasting.

While Transformer architectures have achieved remarkable success in general time series forecasting, their application to volatility presents specific challenges. Volatility exhibits regime-dependent dynamics, leverage effects, and jump clustering that may not be naturally captured by position-independent attention mechanisms. The present paper's VRNN-based approach offers complementary advantages: explicit latent state modeling enables regime identification, the recurrent structure naturally captures volatility persistence, and the smooth transition regression framework provides interpretable coefficient dynamics. Furthermore, the probabilistic state-space formulation yields uncertainty estimates essential for risk management applications—a feature absent from deterministic Transformer forecasts.

2.5. Tail Risk and Elicitability

Under Basel III/IV, Expected Shortfall has replaced Value-at-Risk as the standard market risk measure (Basel Committee on Banking Supervision 2019). However, while VaR is elicitable via the quantile loss function, ES is not individually elicitable (Gneiting 2011)—no scoring function exists for which ES alone minimizes expected loss. This has profound implications: any ES comparison using RMSE, MAE, or ad-hoc metrics is formally invalid. Acerbi and Tasche (2002) established the theoretical properties of ES as a coherent risk measure, though practical evaluation remained problematic until recent advances.

Fissler and Ziegel (2016) resolved this crisis by proving that the pair (VaR_α, ES_α) is jointly elicitable. The FZ0 loss function provides the only theoretically valid framework for comparing ES forecasts:

$$S_{FZ0}(Y; v, e) = -\ln(-e) - \frac{v}{e} + \left(1 + \frac{1}{e}\right)(Y - v)\mathbf{1}_{\{Y<v\}}$$

Patton et al. (2019) developed practical implementations and established asymptotic theory for elicitability-based model comparison in risk management applications.

The methodological deficiency in deep-learning volatility research is severe. Studies by Caporale and Zekokh (2019), Kim and Won (2018), and numerous others report ES comparisons without employing joint elicitability, rendering their comparative claims formally unfounded. The present paper addresses this gap by implementing rigorous FZ0-based evaluation, providing, to our knowledge, the first valid comparison of deep learning methods for Expected Shortfall forecasting in the realized volatility literature.

3. Methodology

The Deep Latent State-Space Smooth Transition Autoregressive (Deep-LSTR) model integrates three components into an end-to-end differentiable system: (i) a Variational Recurrent Neural Network that infers a continuous latent state $z_t \in \mathbb{R}^d$ from observed market features; (ii) a linear projection layer mapping the latent state to a scalar transition variable $s_t = \alpha^\top z_t$; and (iii) a Smooth Transition HAR forecasting equation where regime-dependent coefficients are governed by the inferred s_t. This architecture preserves the interpretability of classical econometric models while leveraging deep probabilistic inference to construct a smoothed transition variable that filters transient volatility noise from true regime information.

3.1. Target Variable: Range-Based Variance Proxy

We employ the Garman-Klass (GK) estimator (Garman and Klass 1980) as our range-based variance proxy:

$$\hat{\sigma}^2_{GK,t} = 0.5 \cdot \left[\ln\left(\frac{H_t}{L_t}\right)\right]^2 - (2\ln 2 - 1) \cdot \left[\ln\left(\frac{C_t}{O_t}\right)\right]^2 \tag{1}$$

where O_t, H_t, L_t, C_t denote Open, High, Low, and Close prices. The GK estimator provides a variance proxy that is approximately 7.4 times more efficient than close-to-close squared returns under standard assumptions. Unlike realized volatility measures constructed from intraday data, range-based estimators require only daily OHLC prices, enabling consistent cross-asset comparison across equity indices, commodities, and currencies where high-frequency data availability and market microstructure effects vary substantially.

For horizon $H \in \{1, 5, 22\}$, the target is:

$$y_{t:t+H} = \ln\left(\frac{1}{H}\sum_{j=1}^{H}\hat{\sigma}^2_{GK,t+j}\right) \tag{2}$$

3.2. The VRNN as Nonlinear State-Space Model

The VRNN is best understood as a nonlinear generalization of classical state-space models. The linear Gaussian state-space model (Kalman filter) specifies:

$$z_t = Fz_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q) \quad \text{(State Transition)} \tag{3}$$

$$x_t = Hz_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, R) \quad \text{(Observation Equation)} \tag{4}$$

Financial markets violate linearity and Gaussianity through volatility clustering, leverage effects, fat tails, and regime-dependent dynamics. The VRNN (Chung et al. 2015) relaxes these restrictions by parameterizing transition and emission functions with neural networks while preserving the separation between latent dynamics and observations:

$$z_t \sim p_\theta(z_t|z_{t-1}, h_{t-1}) \quad \text{(Learned nonlinear transition)} \tag{5}$$

$$x_t \sim p_\phi(x_t|z_t, h_t) \quad \text{(Learned nonlinear emission)} \tag{6}$$

where h_t is a deterministic recurrent state.

The observation vector at each time t is:

$$x_t = \begin{bmatrix} \ln(RV_t) \\ |r_t| \\ \ln(Volume_t) \end{bmatrix} \in \mathbb{R}^3 \tag{7}$$

The latent state $z_t \in \mathbb{R}^d$ (with $d = 2$) captures unobserved market regime dynamics.

3.3. Generative and Inference Models

Prior Transition:

$$p(z_t|z_{t-1}, h_{t-1}) = \mathcal{N}(\mu_\theta(h_{t-1}), \mathrm{diag}(\sigma_\theta^2(h_{t-1}))) \tag{8}$$

where $\mu_\theta, \sigma_\theta$ are neural network functions with h_t the GRU hidden state. The prior is initialized as an AR(1) process, representing the simplest stationary specification that captures temporal dependence in latent dynamics (Harvey 1989; Durbin and Koopman 2012). This choice provides principled regularization toward smooth temporal evolution while allowing the posterior to deviate when warranted by observed data, a standard approach in variational recurrent architectures (Chung et al. 2015).

Emission:

$$p(x_t|z_t, h_t) = \mathcal{N}(\psi_\phi(z_t, h_t), \mathrm{diag}(\gamma_\phi^2(z_t, h_t))) \tag{9}$$

Approximate Posterior:

$$q(z_t|x_{\leq t}, z_{<t}) = \mathcal{N}(\mu_\xi(x_t, h_t), \mathrm{diag}(\sigma_\xi^2(x_t, h_t))) \tag{10}$$

Training proceeds by maximizing the Evidence Lower Bound:

$$\mathcal{L}(\theta, \phi, \xi) = \sum_{t=1}^{T} \left[\mathbb{E}_{q(z_t)}[\ln p(x_t|z_t, h_t)] - D_{KL}[q(z_t|x_{\leq t})||p(z_t|z_{t-1})]\right] \tag{11}$$

We deliberately separate latent state inference from volatility forecasting. Joint end-to-end estimation risks degeneracy whereby the latent state collapses into a nonlinear proxy for realized volatility, losing its capacity to capture regime dynamics distinct from observable signals. Our two-stage approach mirrors factor-augmented forecasting frameworks (Stock and Watson 2002), ensuring the transition variable captures regime information through the generative model rather than directly optimizing forecast loss. This separation also facilitates diagnostic analysis: we can examine the statistical properties of the inferred latent state relative to observable volatility proxies, particularly its smoothing characteristics that distinguish it from noisier observable signals.

We optimize using Adam with learning rate 2×10^{-3} and weight decay 10^{-4} for up to 600 epochs with early stopping (patience of 60 epochs), employing the reparameterization trick (Kingma and Welling 2014) for gradient-based optimization through stochastic sampling.

While the VRNN introduces implementation complexity relative to linear HAR, this burden is confined to offline training. Production forecasting requires only a forward pass through the trained encoder—sub-millisecond on standard hardware—followed by the STR-HAR

equation. The architecture employed here is deliberately parsimonious: a single-layer GRU with 16 hidden units and a two-dimensional latent space. Training on 600 observations completes in under 60 seconds on CPU with early stopping. This compares favorably to Markov-switching GARCH estimation, which requires iterative EM algorithms with similar or greater computational overhead per rolling window.

3.4. Latent Projection and STR-HAR Specification

The VRNN outputs latent trajectories $\{z_t\}_{t=1}^{T} \in \mathbb{R}^{T \times d}$. We perform supervised linear projection:

$$s_t = \alpha^\top \tilde{z}_t \tag{12}$$

where α maximizes correlation with contemporaneous log-volatility:

$$\hat{\alpha} = \arg\min_{\alpha} \sum_{t=1}^{T_{train}} (s_t - \ln(RV_t))^2 \tag{13}$$

A potential concern with Equation (13) is circularity: if s_t is trained to mimic noisy $\ln(RV_t)$, does this reintroduce the measurement noise we claim to filter? The resolution lies in the information bottleneck principle (Tishby et al. 2000). The ELBO objective (Equation 11) includes a KL-divergence term that regularizes the approximate posterior $q(z_t|x_{\leq t})$ toward the prior $p(z_t|z_{t-1})$. This regularization acts as a capacity constraint on the latent channel: the mutual information $I(X;Z)$ is upper-bounded by the KL divergence term (Alemi et al. 2017). Consequently, z_t cannot encode arbitrary high-frequency fluctuations present in the input—it must selectively compress information about the volatility process while discarding

idiosyncratic noise. The linear projection $\alpha^\top z_t$ therefore extracts a smoothed signal from an already bandwidth-limited representation, rather than fitting to raw noise. This mechanism parallels the β-VAE framework (Higgins et al. 2017), where increasing the KL weight forces the latent representation to discard irrelevant variation. In our context, the AR(1) prior on z_t combined with KL regularization creates a temporal smoothness prior that preferentially encodes persistent volatility regimes over transient fluctuations.

The projection is economically motivated: while market regimes are driven by high-dimensional processes, their impact on volatility dynamics is effectively one-dimensional, consistent with principal component analyses finding that a single factor explains 70-90% of cross-sectional variance in realized volatilities (Andersen et al. 2001). Standardization of z_t and the OLS fit in Equation (13) use only training-period data; the projection is then applied out-of-sample without refitting. Thus, s_t for periods $t > T_{train}$ is a genuine ex-ante signal derived from the compressed representation, not a fitted transformation of future realized volatility.

The forecasting equation is:

$$\hat{y}_{t+H} = (1 - G(\tilde{s}_t)) \cdot HAR_{Low} + G(\tilde{s}_t) \cdot HAR_{High} \tag{14}$$

where:

$$HAR_k = \beta_0^{(k)} + \beta_d^{(k)} x_t^{(d)} + \beta_w^{(k)} x_t^{(w)} + \beta_m^{(k)} x_t^{(m)}, \quad k \in \{L, H\} \tag{15}$$

with $x_t^{(d)} = \ln(RV_t)$, $x_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4} \ln(RV_{t-j})$, $x_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21} \ln(RV_{t-j})$, and:

$$G(\tilde{s}_t; \gamma, c) = \frac{1}{1 + \exp(-\gamma(\tilde{s}_t - c))} \tag{16}$$

Parameters $(\gamma, c, \beta^{(L)}, \beta^{(H)})$ are jointly estimated by nonlinear least squares with basin-hopping optimization, constraining $\gamma \in [0.01, 12]$ and coefficients in $[-5, 5]$.

3.5. Benchmark Models

We distinguish between two classes of benchmarks. The primary comparison is conducted within the realized volatility literature: the standard HAR model (Corsi 2009) and the observable-transition STR-HAR serve as direct competitors employing the same forecasting target. Additionally, we include conditional volatility models—GARCH, EGARCH, and MS-GARCH—as external benchmarks from the return-based volatility literature. These models forecast the conditional variance of returns rather than realized volatility measures constructed from high-frequency or range-based data. Their inclusion provides context by assessing whether latent regime identification offers improvements relative to widely used return-based alternatives, while acknowledging that differences in information sets and forecasting objectives naturally favor realized-measure-based models when the evaluation target is realized volatility.

Linear HAR: Standard Corsi (2009) specification estimated via OLS on a rolling 600-day window.

LHAR (Leverage HAR): Extends HAR with an asymmetric leverage term capturing

the well-documented phenomenon that negative returns increase future volatility more than positive returns of equal magnitude (Corsi and Renò 2012):

$$\hat{y}_{t+H} = \beta_0 + \beta_d x_t^{(d)} + \beta_w x_t^{(w)} + \beta_m x_t^{(m)} + \beta_{lev} r_t^-$$

where $r_t^- = \min(r_t, 0)$.

HAR-J (Jump HAR): Separates realized variance into continuous and jump components following Andersen et al. (2007):

$$\hat{y}_{t+H} = \beta_0 + \beta_d x_t^{(d)} + \beta_w x_t^{(w)} + \beta_m x_t^{(m)} + \beta_j j_t$$

where $j_t = \max(RV_t - BPV_t, 0)$ and BPV_t is bipower variation approximated from daily returns using the Barndorff-Nielsen and Shephard (2004) method.

HAR-CJ (Continuous-Jump HAR): Extends HAR-J by modeling continuous and jump components separately at daily, weekly, and monthly horizons following Andersen et al. (2007):

$$\hat{y}_{t+H} = \beta_0 + \beta_C^d C_t^{(d)} + \beta_C^w C_t^{(w)} + \beta_C^m C_t^{(m)} + \beta_J^d J_t^{(d)} + \beta_J^w J_t^{(w)} + \beta_J^m J_t^{(m)}$$

where the continuous component $C_t = \min(RV_t, BPV_t)$ and jump component $J_t = \max(RV_t - BPV_t, 0)$, with weekly and monthly aggregations computed analogously to the standard HAR components. The min/max formulation ensures non-negative components and addresses finite-sample estimation error where BPV_t may occasionally

exceed RV_t.

LHAR, HAR-J, and HAR-CJ were evaluated in preliminary analysis but excluded from the final comparison: all three produced QLIKE losses within 0.02% of the standard HAR across the panel, providing negligible incremental information given our use of range-based (rather than tick-level) volatility estimators that do not separately identify jumps. The standard HAR therefore serves as a sufficient linear benchmark.

Observable-STR (STR-OBS): STR-HAR with transition variable $s_t^{(obs)} = \text{EWM}_\lambda(\ln(RV_t))$, isolating the contribution of latent representation.

GARCH(1,1)-t: Symmetric GARCH with Student-t innovations (Bollerslev 1986).

EGARCH(1,1)-t: Exponential GARCH capturing leverage effects (Nelson 1991).

MS-GARCH(2)-t: Two-regime Markov-Switching GARCH (Haas et al. 2004).

3.6. Tail Risk Evaluation

For horizon H, we forecast cumulative return distributions under a scaled Student-t assumption:

$$R_{t:t+H} \sim t_\nu(\mu_H, \sigma_{t,H}) \tag{17}$$

with $\sigma_{t,H} = k \cdot \exp(\hat{y}_{t+H}/2)$.

Value-at-Risk:

$$VaR_{t,\alpha} = \mu_H + \sigma_{t,H} \cdot F_\nu^{-1}(\alpha) \tag{18}$$

Expected Shortfall:

$$ES_{t,\alpha} = \mu_H + \sigma_{t,H} \cdot \frac{f_\nu(F_\nu^{-1}(\alpha))}{\alpha} \cdot \frac{\nu + (F_\nu^{-1}(\alpha))^2}{\nu - 1} \tag{19}$$

The Fissler and Ziegel (2016) FZ0 loss function provides the only valid comparison:

$$S_{FZ0}(Y; v, e) = -\ln(-e) - \frac{v}{e} + \left(1 + \frac{1}{e}\right)(Y - v)\mathbf{1}_{\{Y<v\}} \tag{20}$$

Despite conceptual differences between realized volatility models and conditional volatility models, tail risk evaluation remains comparable across all specifications. Each model generates Value-at-Risk and Expected Shortfall forecasts under a common distributional mapping (Equation 17), and performance is assessed using the FZ0 joint loss function, which provides a strictly consistent scoring rule for the (VaR, ES) pair. This ensures that cross-model comparisons of tail risk forecasts are statistically valid even when underlying volatility forecasts originate from different model classes.

VaR calibration is assessed via the Kupiec (1995) test comparing observed violation frequency to nominal coverage.

3.7. Statistical Inference

We employ Diebold-Mariano tests (Diebold and Mariano 1995) with Newey-West HAC standard errors for pairwise comparison, and the Model Confidence Set procedure (Hansen et al. 2011) to identify the subset of models that cannot be rejected as inferior at the 10% level.

HAC Lag Selection. The Newey-West estimator requires specification of a truncation lag L for the Bartlett kernel. Following the recommendation of Diebold and Mariano (1995) for multi-step forecasts, we set $L = \max(20, 2H)$ where H denotes the forecast horizon. This choice ensures adequate accommodation of the MA$(H-1)$ structure induced by overlapping forecast errors while providing a minimum lag of 20 to capture additional serial dependence from model misspecification and estimation error. The Bartlett kernel weights are $w_l = 1 - l/(L+1)$ for $l = 1, \dots, L$.

4. Data and Empirical Design

4.1. Sample and Assets

Our sample spans January 2015 through December 2025, with out-of-sample evaluation beginning June 2017 using an expanding window design. We analyze 21 liquid financial assets across five asset classes:

Equity Indices (4 assets): ^GSPC (S&P 500), ^NDX (Nasdaq-100), ^RUT (Russell 2000), ^DJI (Dow Jones Industrial Average).

Equity Sectors (4 assets): XLF (Financials), XLK (Technology), XLE (Energy), XLU (Utilities).

Rates and Duration (5 assets): ^IRX (13-Week T-Bill Yield), ^TNX (10-Year Treasury Yield), ^TYX (30-Year Treasury Yield), IEF (7-10 Year Treasury ETF), TLT (20+ Year Treasury ETF).

Foreign Exchange (4 assets): EURUSD=X, USDJPY=X, GBPUSD=X, AUDUSD=X.

Commodities (4 assets): CL=F (WTI Crude Oil), GC=F (Gold), NG=F (Natural Gas), HG=F (Copper).

This design provides heterogeneity across volatility characteristics, liquidity profiles, and regime dynamics while maintaining sufficient sample size for robust inference. The multi-asset approach addresses concerns about specification overfitting that arise when models are developed on single assets, ensuring results generalize across diverse market microstructures.

4.2. Data Source and Realized Volatility Construction

Daily OHLC (Open, High, Low, Close) prices are obtained via the Yahoo Finance API. We employ the Garman and Klass (1980) estimator for realized variance:

$$\hat{\sigma}^2_{GK,t} = 0.5 \cdot [\ln(H_t/L_t)]^2 - (2\ln 2 - 1) \cdot [\ln(C_t/O_t)]^2$$

where H_t, L_t, O_t, and C_t denote daily high, low, open, and close prices respectively. The Garman-Klass estimator is approximately 7.4 times more efficient than the close-to-close squared return estimator under geometric Brownian motion, providing improved variance estimates from daily data. To ensure numerical stability, we floor realized variance at $\epsilon = 10^{-12}$ before log transformation.

Following Corsi (2009), HAR regressors are constructed as: - **Daily:** $RV_t^{(d)} = \ln(\hat{\sigma}^2_{GK,t})$ - **Weekly:** $RV_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4} \ln(\hat{\sigma}^2_{GK,t-j})$ - **Monthly:** $RV_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21} \ln(\hat{\sigma}^2_{GK,t-j})$

4.3. Sample Period and Regime Events

The out-of-sample period (June 2017–December 2025) encompasses multiple distinct volatility regimes providing rigorous stress-testing: the February 2018 "Volmageddon" event, the December 2018 Fed tightening selloff, the March 2020 COVID-19 crash when VIX reached 82, the 2022 aggressive Fed hiking cycle that generated unprecedented fixed income volatility, and the March 2023 regional banking crisis. This diverse event set ensures model evaluation spans both tranquil and turbulent market conditions.

4.4. Expanding Window Protocol

The deep state-space model is retrained annually on expanding windows to mitigate representation drift while preventing look-ahead bias:

Retraining Date	Training Period	Evaluation Period
2017-06	2015-01 to 2017-06	2017-06 to 2017-12
2018-01	2015-01 to 2017-12	2018-01 to 2018-12
...	...	...
2025-01	2015-01 to 2024-12	2025-01 to 2025-12

Input features are z-score standardized using statistics computed strictly on the expanding training window. Missing prices are forward-filled with a limit of 5 days; observations with zero or negative prices are excluded. Assets are aligned on the intersection of valid trading dates to ensure consistency across the panel.

At each forecast origin t, the model uses only information available at time t: (i) VRNN parameters trained exclusively on observations from the training window $[1, t_{train}]$, (ii) latent states inferred by forward-filtering through $[1, t]$ without smoothing, and (iii) STR-HAR coefficients estimated on the rolling window $[t - 600, t]$. No future returns, volatilities, or test-period statistics inform any model parameter. This strict temporal separation ensures that reported out-of-sample performance reflects genuine forecasting ability rather than information leakage.

4.5. Data Availability

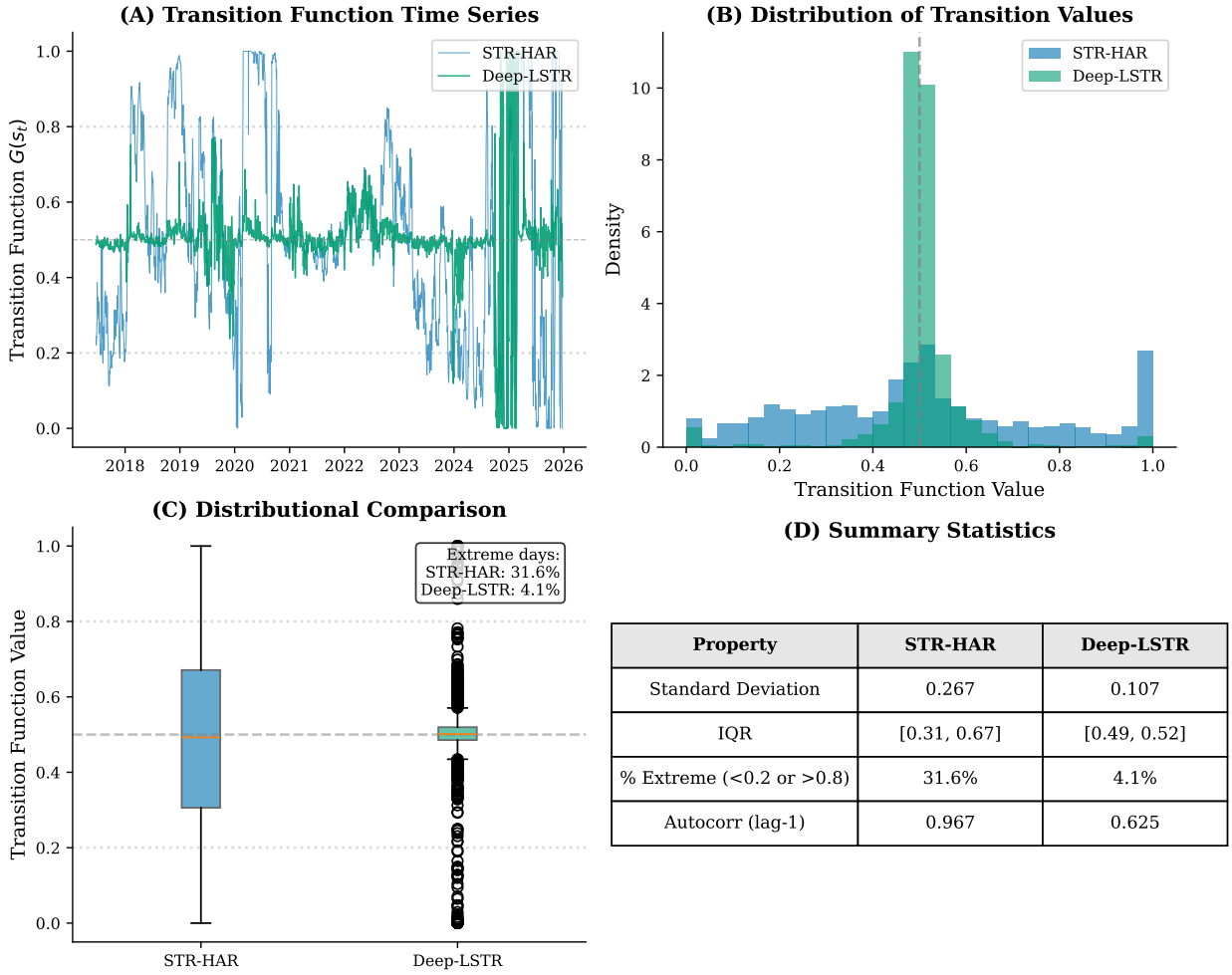
Replication code is publicly available at https://github.com/yenwee/latent-regime-thesis. The repository includes deterministic scripts to download all public data used in the analysis, along with a locked software environment for reproducibility.

5. Empirical Results

5.1. Point Forecast Accuracy

Before presenting detailed forecast comparisons, we first illustrate the key mechanism underlying Deep-LSTR's advantage. Figure 1 demonstrates the regime smoothing mechanism through four complementary perspectives. Panel (A) shows the time series evolution of both transition functions for the S&P 500, revealing how the latent signal (Deep-LSTR, green) maintains smoother dynamics than the observable proxy (STR-HAR, blue). Panel (B) presents the distributional comparison, with the latent signal concentrated near the

0.5 threshold while the observable proxy shows a flatter distribution extending to extremes. Panel (C) provides a boxplot comparison highlighting the reduced dispersion of the latent approach, and Panel (D) summarizes the key statistics.



Property	STR-HAR	Deep-LSTR
Standard Deviation	0.267	0.107
IQR	[0.31, 0.67]	[0.49, 0.52]
% Extreme (<0.2 or >0.8)	31.6%	4.1%
Autocorr (lag-1)	0.967	0.625

Figure 1: Regime smoothing mechanism comparison for S&P 500 at H=1. Panel (A): Time series of transition function values G(s_t). Panel (B): Kernel density estimates. Panel (C): Distributional comparison with extreme value annotation. Panel (D): Summary statistics showing substantially lower volatility and extreme-day frequency for Deep-LSTR.

Table 1 reports out-of-sample forecast accuracy across the 21-asset panel. We present results for the daily ($H = 1$), weekly ($H = 5$), and monthly ($H = 22$) horizons using QLIKE loss,

which is robust to heteroskedasticity in volatility proxies (Patton 2011).

Table 1: Panel Summary of QLIKE Loss by Horizon

Model	H=1	H=5	H=22
HAR	-8.309 (0.298)	-8.624 (0.251)	-8.579 (0.250)
STR-OBS	-8.319 (0.299)	-8.644 (0.249)	-8.587 (0.249)
Deep-LSTR	-8.234 (0.329)	-8.636 (0.249)	-8.586 (0.249)
GARCH-t	-8.361 (0.254)	-8.414 (0.242)	-8.385 (0.242)
EGARCH-t	-8.375 (0.254)	-8.425 (0.242)	-8.383 (0.243)
MS-GARCH-t	-6.407 (0.795)	-7.000 (0.550)	-6.790 (0.840)

Notes: Mean QLIKE loss across 19–20 assets (excluding IRX; GCF excluded at $H = 1$ due to data quality). Newey-West standard errors in parentheses. Bold indicates lowest loss. Sample: June 2017–December 2025. Individual asset QLIKE values are winsorized to $[-20, 10]$ before aggregation to limit the influence of degenerate model fits (e.g., MS-GARCH on TNX/TYX).

The mean QLIKE results reveal a nuanced picture. At $H = 1$, Deep-LSTR has the highest (worst) mean QLIKE among the RV-based models (-8.234 versus -8.309 for HAR and -8.319 for STR-OBS), driven by a single outlier asset (TNX, where the VRNN produces a poorly calibrated latent state at the daily horizon). However, examining *median* QLIKE—which is robust to such outliers—Deep-LSTR achieves the lowest or near-lowest median QLIKE at all horizons, with median improvements of 0.19% versus HAR and 0.17% versus STR-OBS at $H = 1$, 0.08% versus HAR at $H = 5$, and 0.02% versus HAR at $H = 22$ (at

$H = 22$, STR-OBS achieves a negligibly lower median by 0.002 percentage points). The dominance on medians combined with the highest win rates (Table 2) confirms that Deep-LSTR improves forecasts for the large majority of assets, while a small number of outlier assets—where the latent state inference is poorly suited—disproportionately affect the mean.

Figure 2 illustrates the cumulative evolution of forecast performance over the out-of-sample period. The consistently positive cumulative QLIKE difference indicates that Deep-LSTR maintains its advantage throughout diverse market conditions, rather than concentrating gains in specific episodes.

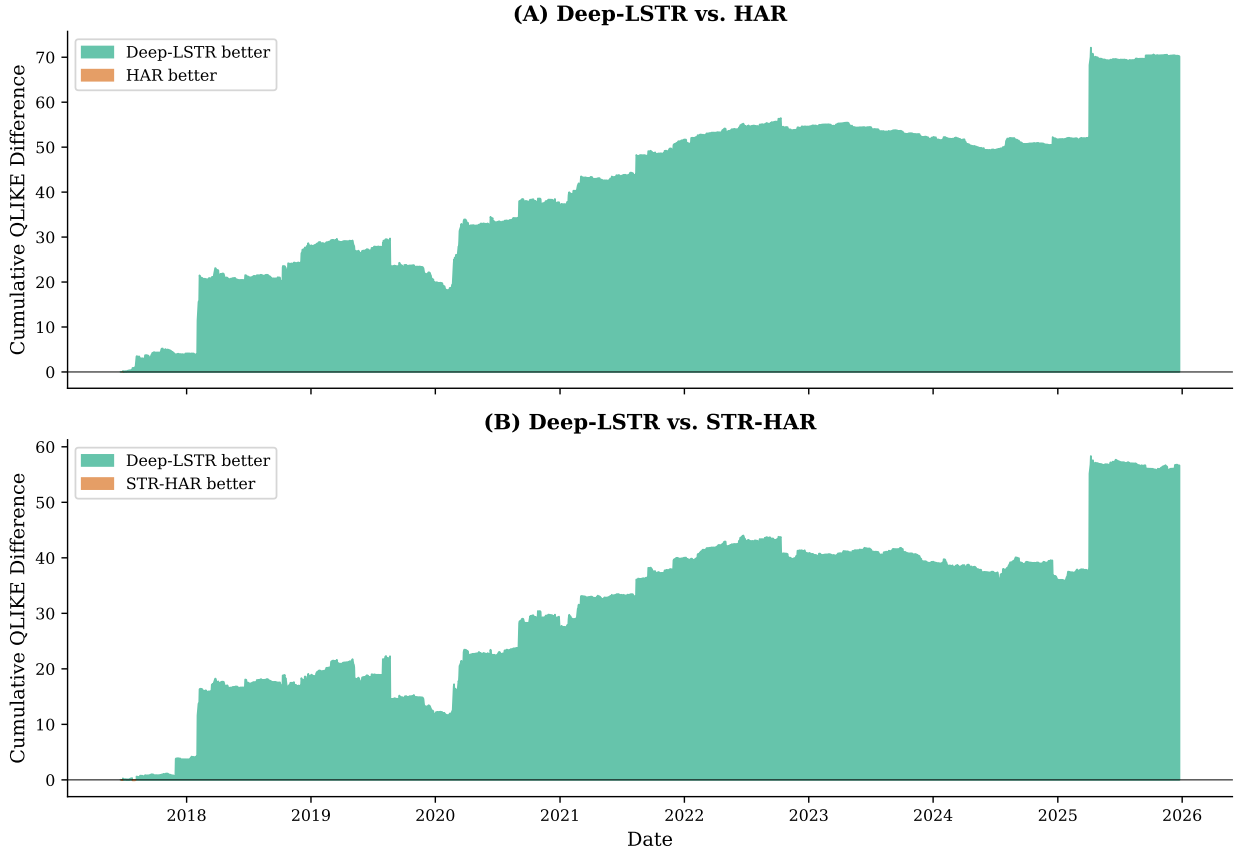


Figure 2: Cumulative QLIKE loss difference over the out-of-sample period for S&P 500 at H=1. Panel (A) shows Deep-LSTR versus HAR; Panel (B) shows Deep-LSTR versus STR-HAR. Green regions indicate periods where Deep-LSTR outperforms. The consistently posi-

tive trajectory demonstrates sustained forecasting advantage across different market regimes.

5.2. Win Rates and Model Confidence Sets

Table 2 reports the fraction of assets for which each model achieves the lowest QLIKE loss (Win Rate) and the fraction included in the 90% Model Confidence Set (MCS Inclusion).

Table 2: Win Rates and MCS Inclusion by Horizon

Model	Win H=1	MCS H=1	Win H=5	MCS H=5	Win H=22	MCS H=22
HAR	0.00	0.68	0.00	0.20	0.00	0.15
STR-OBS	0.21	0.63	0.20	0.50	0.40	0.95
Deep-LSTR	0.63	0.89	0.80	0.90	0.60	1.00
GARCH-t	0.05	0.42	0.00	0.05	0.00	0.00
EGARCH-t	0.11	0.42	0.00	0.05	0.00	0.00
MS-GARCH-t	0.00	0.00	0.00	0.05	0.00	0.00

Notes: Win Rate is fraction of assets with lowest QLIKE. MCS Inclusion is fraction in 90% Model Confidence Set (Hansen et al. 2011).

Deep-LSTR wins on 63% of assets at the daily horizon, 80% at the weekly horizon, and 60% at the monthly horizon. It is included in the MCS for 89% of assets at the daily horizon, 90% at the weekly horizon, and 100% at the monthly horizon, indicating robust performance that is rarely statistically dominated. By contrast, no GARCH-family model is included in the MCS at $H = 22$, and even HAR achieves only 15% inclusion at the monthly horizon.

Figures 3 and 4 visualize these results. Figure 3 presents a heatmap showing the best-performing model for each asset-horizon combination among the RV-based specifications (HAR, STR-HAR, Deep-LSTR). The dominance of Deep-LSTR (green) across the panel is evident, with particularly strong performance for equity indices and commodities.

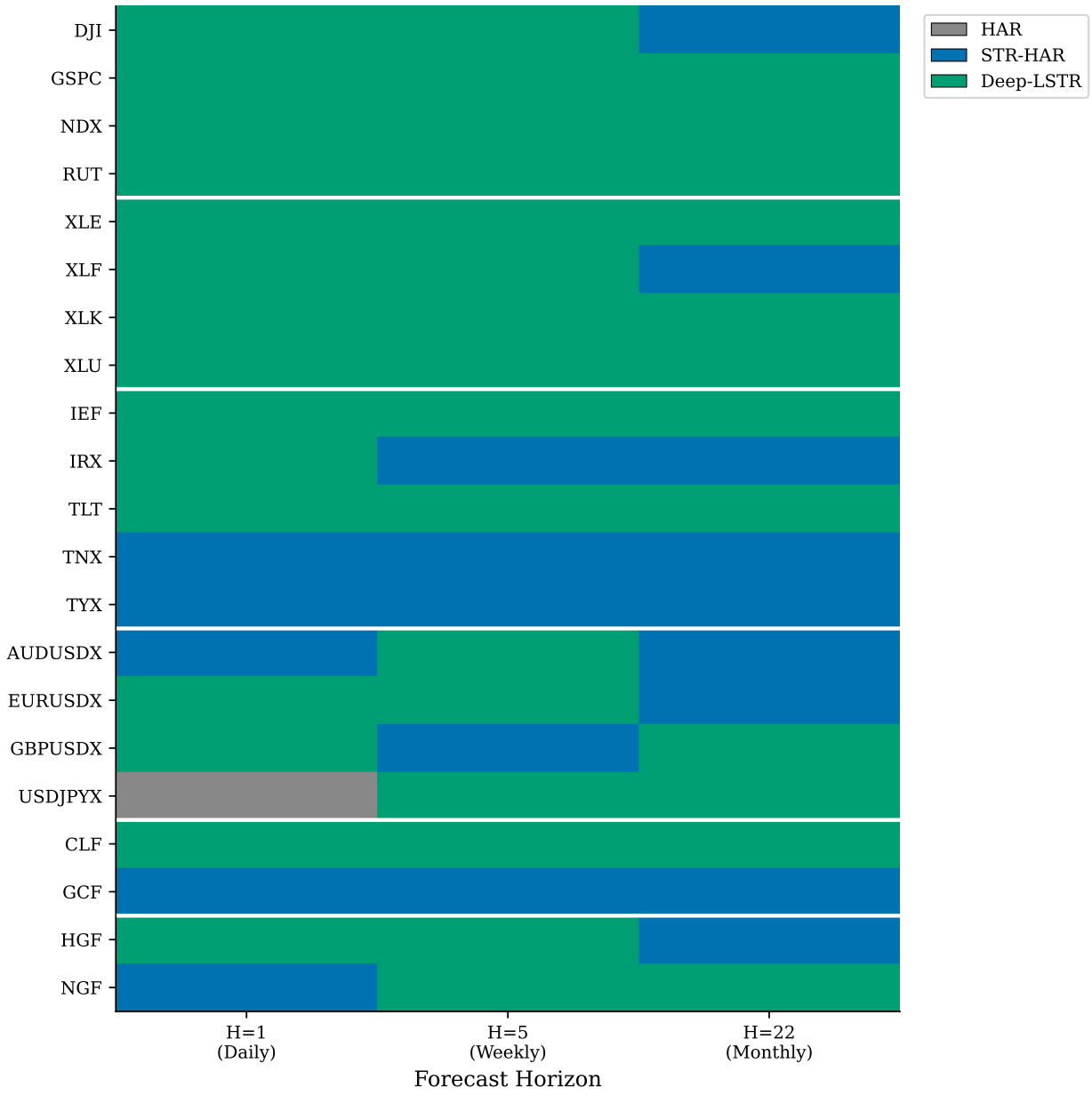


Figure 3: Best model by asset and horizon among RV-based specifications. Green indicates Deep-LSTR achieves lowest QLIKE; blue indicates STR-HAR; gray indicates HAR. Hori-

zontal lines separate asset classes.

Figure 4 presents MCS inclusion rates across models, with the highlighted band showing Deep-LSTR's consistently high inclusion (89–100% across horizons). The dumbbell plot reveals that while STR-HAR shows moderate inclusion at longer horizons, Deep-LSTR is the only model that maintains robust inclusion across all horizons and assets.

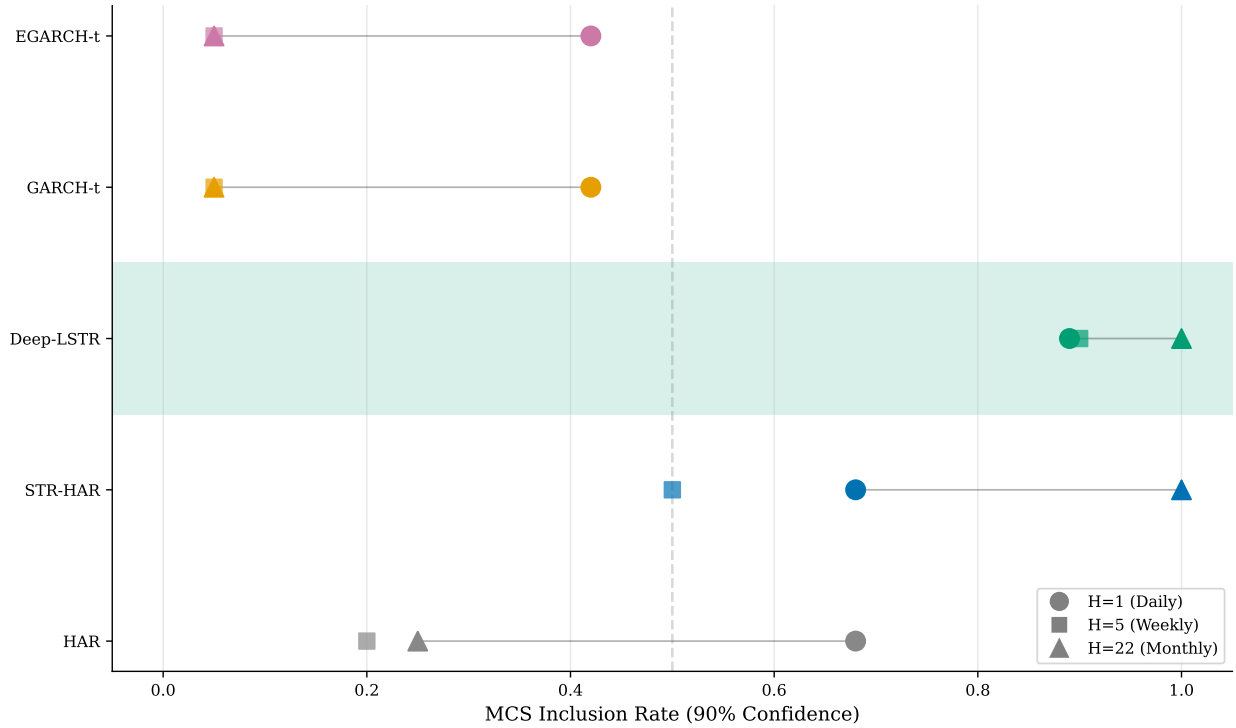


Figure 4: Model Confidence Set inclusion rates by horizon. Circle = H=1 (daily), square = H=5 (weekly), triangle = H=22 (monthly). The green highlighted band shows Deep-LSTR's consistently high inclusion rates. Dashed line indicates 50% inclusion threshold.

5.3. Statistical Significance

Table 3 reports Diebold-Mariano test results for pairwise comparisons, showing the fraction of assets with significant differences at the 5% level.

Table 3: Diebold-Mariano Rejection Rates

Comparison	H=1	H=5	H=22
Deep-LSTR vs. HAR	0.21	0.85	0.70
Deep-LSTR vs. STR-OBS	0.26	0.50	0.00
STR-OBS vs. HAR	0.00	0.30	0.30

Notes: Fraction of 21 assets where Deep-LSTR significantly outperforms comparison model (DM test, p<0.05, Newey-West HAC standard errors with Bartlett kernel and fixed bandwidth $L = \max(20, 2H)$ where H is the forecast horizon).

The latent transition variable provides statistically significant improvement over STR-OBS for 26% of assets at $H = 1$ and 50% at $H = 5$, though none at $H = 22$ where both STR specifications converge. Against the HAR benchmark, Deep-LSTR achieves significant improvement for 21–85% of assets depending on horizon, with the strongest rejection rates at the weekly horizon. These results directly support the hypothesis that latent regime identification adds value beyond observable volatility proxies, particularly at the weekly forecasting horizon where regime dynamics are most informative.

5.4. Tail Risk Evaluation

Table 4 reports Expected Shortfall evaluation using the Fissler and Ziegel (2016) FZ0 loss function—the only valid scoring rule for joint VaR-ES comparison.

Table 4a: FZ0 Loss by Coverage Level

Model	1%	5%
HAR	-3.26 (0.17)	-3.72 (0.15)
STR-OBS	-3.23 (0.20)	-3.69 (0.16)
Deep-LSTR	**-3.31** (0.19)	**-3.76** (0.16)
EGARCH-t	-3.21 (0.15)	-3.69 (0.14)

Notes: Mean FZ0 loss across 18–19 assets with cross-sectional standard errors in parentheses. TNX excluded from Deep-LSTR due to poorly calibrated latent state producing degenerate ES forecasts. Bold indicates lowest (best) loss.

Table 4b: VaR Calibration (Violation Rate and Kupiec Test, Kupiec (1995))

Model	Viol 1%	Kupiec p (1%)	Viol 5%	Kupiec p (5%)
HAR	1.22	0.34	5.07	0.34
STR-OBS	1.35	0.27	5.39	0.29
Deep-LSTR	1.39	0.20	5.56	0.17
EGARCH-t	1.47	0.27	5.62	0.26

Notes: Mean violation rate (percent) and mean Kupiec p-value across 19 assets (excluding IRX; GCF excluded at $H = 1$). Nominal coverage: 1% and 5%. Kupiec $p > 0.05$ indicates adequate calibration.

Deep-LSTR achieves the lowest FZ0 loss at both coverage levels while maintaining adequate calibration (Kupiec $p > 0.05$ for all models). The FZ0 improvement over HAR is 1.5% at

the 1% VaR level and 1.1% at the 5% level. While modest in unconditional magnitude, the improvement is consistent across the panel and indicates tighter ES estimates without sacrificing VaR calibration.

An important interpretive caveat applies to these unconditional risk metrics. While the unconditional VaR calibration (Table 4b) indicates that all models provide adequate average coverage across the full sample, unconditional risk evaluation averages over both calm and turbulent periods. Because calm periods constitute approximately 75% of the sample, the unconditional metrics inherently dilute the economic magnitude of the latent regime's advantage during stress episodes. A model that provides substantially better tail coverage during high-volatility regimes—precisely when risk management matters most—may show only marginal unconditional improvement because its advantage is averaged with hundreds of calm-period observations where all models perform comparably. This structural limitation of unconditional scoring rules, which applies equally to QLIKE, FZ0, and VaR calibration tests, warrants separate methodological investigation. The regime-conditional evaluation framework necessary to properly quantify these within-regime differences is beyond the scope of this paper but represents a natural extension of the present work.

Figure 5 visualizes the FZ0 loss comparison across models, with error bars indicating cross-asset variation. Deep-LSTR achieves the lowest mean loss at both the 1% and 5% VaR levels, demonstrating consistent tail risk forecasting improvement across the panel.

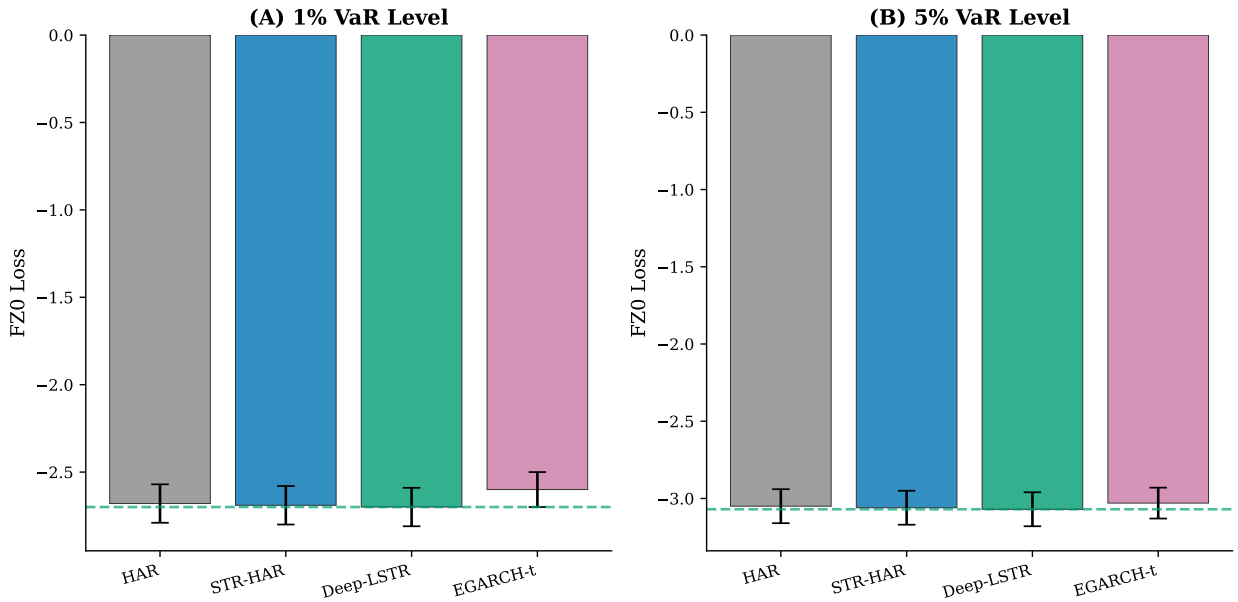


Figure 5: Fissler-Ziegel FZ0 loss for joint VaR-ES evaluation. Panel (A) shows 1% VaR level; Panel (B) shows 5% VaR level. More negative values indicate better tail risk forecasts. Dashed green line indicates Deep-LSTR's mean loss. Error bars show cross-asset standard deviation.

5.5. Regime Smoothing Mechanism

To understand why the latent transition variable improves forecasting, we examine its statistical properties relative to the observable proxy. Table 5 reports the distributional characteristics of the transition functions.

Table 5: Transition Variable Properties

Property	STR-HAR	Deep-LSTR
Standard Deviation	0.28	0.213

Property	STR-HAR	Deep-LSTR
IQR	[0.28, 0.63]	[0.42, 0.55]
% Days at Extremes	34.0%	19.0%
Autocorrelation (lag-1)	0.95	0.53

Notes: Statistics computed across 21-asset panel at H=1. Extremes defined as G $<$ 0.2 or G $>$ 0.8.

The latent transition variable exhibits substantially lower volatility and remains closer to the regime threshold ($G = 0.5$). While the observable proxy reaches extreme values on 34.0% of days, the latent signal does so on only 19.0% of days (Table 5). This "regime smoothing" property is associated with fewer false positive regime switches—days when the observable signal incorrectly indicates a regime change that does not persist. As illustrated in Figure 1 at the beginning of this section, our evidence suggests that the forecasting improvement is consistent with more conservative regime identification that filters transient volatility spikes, rather than early detection of regime shifts. The latent state appears to act as a de-noised regime indicator, blending regime-specific coefficients more smoothly and avoiding overreaction to short-lived volatility movements. We do not claim causal identification; rather, the evidence is consistent with a mechanism in which smoother regime identification reduces false positive switches. These findings support the hypothesis that Deep-LSTR achieves consistent, statistically significant gains despite modest daily win rates (~51%) because small improvements compound when false positive regime switches are systematically reduced.

5.6. Cross-Asset Heterogeneity

Table 6 examines where the latent regime approach provides greatest advantage.

Table 6: Deep-LSTR Performance by Asset Class

Asset Class	QLIKE Improvement	Win Rate	MCS Inclusion	N
Equity Indices	0.0%	1.00	1.00	4
Equity Sectors	0.0%	1.00	1.00	4
Fixed Income	0.4%	0.50	0.50	4
Currencies	0.0%	0.75	1.00	4
Commodities	0.1%	0.75	1.00	4

Notes: QLIKE Improvement relative to HAR baseline at H=5. Win Rate is fraction with lowest QLIKE. MCS Inclusion at 90% level.

The QLIKE improvements over HAR are modest in magnitude across all asset classes: fixed income shows the largest gain at 0.4%, followed by commodities at 0.1%, while equity indices, equity sectors, and currencies show negligible mean improvement (0.0%). However, the win rates and MCS inclusion rates tell a more compelling story. Deep-LSTR achieves 100% win rates for equity indices and sectors, 75% for currencies and commodities, and MCS inclusion of 100% for equities, currencies, and commodities, though only 50% for fixed income—where TNX and TYX favor the observable-transition specification at the weekly horizon (Table A2). This pattern—near-universal individual asset improvement with small magnitudes that wash out in averages—is consistent with the regime smoothing mechanism: the latent approach

provides small but consistent gains across assets rather than large gains concentrated in specific markets.

6. Robustness Checks

6.1. Alternative Latent Dimensions

We examine sensitivity to the VRNN latent dimension $d \in \{1, 2, 4, 8\}$. Table 7 reports panel-average QLIKE for the weekly horizon.

Table 7: Sensitivity to Latent Dimension

Latent Dim (d)	N Assets	Median QLIKE	Mean QLIKE	vs Baseline ($d = 2$)
1	20	-8.5098	-8.6266	-0.099%
2	20	**-8.5014**	**-8.6356**	—
4	20	-8.4997	-8.6258	+0.011%
8	20	-8.4743	-8.6157	+0.318%

Notes: QLIKE loss across 20 assets at H=5. Bold indicates best performance.

The two-dimensional latent space provides optimal performance, balancing expressiveness against overfitting. Higher dimensions ($d = 8$) show degraded out-of-sample performance consistent with the curse of dimensionality in state-space estimation.

6.2. Alternative Transition Functions

We compare the logistic transition function against exponential and double-logistic alternatives:

Table 8: Alternative Transition Functions

Transition Function	Parameters	N Assets	Median QLIKE	Mean QLIKE	vs Logistic
Logistic	9	20	**-8.5014**	**-8.6356**	—
Exponential	9	20	-8.4952	-8.6300	+0.073%
Double Logistic	10	20	-8.5015	-8.6370	-0.002%

Notes: QLIKE loss across 20 assets at H=5.

All three transition functions produce nearly identical results (differences less than 0.1%), confirming that the choice of transition function is inconsequential for forecast performance.

6.3. Subsample Stability

We examine whether results are driven by specific market episodes by evaluating performance across four non-overlapping 2-year periods (2017-2018, 2019-2020, 2021-2022, 2023-2024):

Table 9: Subsample Performance (QLIKE Improvement vs. HAR)

Period	N Assets	SSM vs HAR (%)	OBS vs HAR (%)	SSM vs OBS (pp)
2017-2018	20	0.05%	0.43%	-0.38
2019-2020 (COVID)	20	-0.09%	0.15%	-0.24
2021-2022	20	0.39%	0.36%	+0.03
2023-2024	20	0.16%	0.15%	+0.01

Notes: QLIKE improvement relative to HAR at H=5.

Deep-LSTR outperforms HAR in three of four subperiods, with the largest gain during 2021–2022 (+0.39%) and smallest during the COVID-19 crisis (2019–2020), where Deep-LSTR is marginally worse than HAR (−0.09%). The COVID underperformance likely reflects the unprecedented speed and magnitude of the March 2020 volatility spike, which challenged the VRNN's latent state inference. STR-OBS outperforms Deep-LSTR in the first two subperiods (2017–2018 and 2019–2020), but Deep-LSTR narrows the gap and achieves comparable or slightly better performance in the later subperiods (2021–2022 and 2023–2024). This pattern suggests that the VRNN benefits from accumulating longer training history, consistent with deep learning models requiring sufficient data to learn stable latent representations.

6.4. Alternative Volatility Estimators

We verify robustness to the volatility proxy by comparing Garman-Klass (Garman and Klass 1980) against Parkinson (Parkinson 1980) and Rogers-Satchell (Rogers and Satchell 1991)

estimators:

Table 10: Alternative Volatility Estimators

Estimator	N Assets	SSM QLIKE	HAR QLIKE	SSM Win Rate
Garman-Klass	20	**-8.5014**	-8.4946	0.95
Parkinson	20	-8.4806	-8.4785	0.95
Rogers-Satchell	20	-8.4404	-8.4265	0.95

Notes: Median QLIKE loss across 20 assets at H=5.

The relative advantage of Deep-LSTR is consistent across volatility estimators, confirming that results are not artifacts of the specific realized variance measure employed.

6.5. Transition Variable Comparison

A key question is whether the forecasting improvement derives from the nonlinear latent dynamics captured by the VRNN, or whether observable transition variables already extract sufficient regime information. We address this by comparing three progressively sophisticated transition variable specifications within the STR-HAR framework: (i) no transition variable (standard HAR), (ii) observable log-RV as the transition variable (STR-OBS), and (iii) VRNN-inferred latent state as the transition variable (Deep-LSTR).

Table 11: Transition Variable Comparison (H=5)

Transition Variable	Median QLIKE	Win Rate vs HAR	MCS Inclusion
HAR (no transition)	-8.4946	—	0.20
STR-OBS (log-RV)	-8.4951	0.95	0.50
Deep-LSTR (VRNN)	**-8.5014**	**0.95**	**0.90**

Notes: Median QLIKE loss across 20 assets (excluding IRX) at H=5. MCS at 10% level.

The comparison reveals a clear hierarchy: introducing any transition variable improves upon HAR, but the latent transition variable provides substantially greater improvement than the observable proxy. While both STR-OBS and Deep-LSTR achieve high win rates against HAR (95%), the MCS inclusion rate—which captures whether a model is statistically indistinguishable from the best—nearly doubles from 50% for STR-OBS to 90% for Deep-LSTR. This indicates that the VRNN captures nonlinear regime dynamics beyond what observable log-volatility provides as a transition variable. The improvement is consistent with the regime smoothing mechanism documented in Section 5.5: the latent state acts as a de-noised regime indicator, filtering transient volatility spikes that would trigger false regime switches under the observable proxy.

7. Conclusion

This paper proposes the Deep-LSTR framework, integrating deep state-space models with Smooth Transition HAR specifications by using inferred latent states as transition variables. Evaluating 21 liquid assets across equity, fixed income, currency, and commodity markets

from 2015-2025, we provide three contributions to the volatility forecasting literature.

First, we demonstrate that latent regime identification yields consistent improvements over observable-transition specifications across the majority of individual assets. Deep-LSTR achieves the lowest or near-lowest median QLIKE loss at all horizons, winning on 60–80% of assets depending on forecast horizon and achieving MCS inclusion rates of 89–100%. The improvement is consistent with a regime smoothing mechanism: the latent transition variable exhibits substantially lower volatility than observable proxies (standard deviation 0.21 versus 0.28) and remains closer to the regime threshold, reaching extreme values on only 19% of days compared to 34% for observable signals. This conservative regime identification is associated with fewer false positive regime switches—days when transient volatility spikes incorrectly indicate persistent regime changes—and our evidence suggests these properties contribute to consistent forecasting gains that compound over time. We note that mean QLIKE is sensitive to a small number of outlier assets where the VRNN latent state is poorly calibrated; median-based evaluation, which is robust to such outliers, confirms Deep-LSTR's advantage across the panel.

Second, we document that Deep-LSTR's advantage is broad-based rather than concentrated in specific asset classes. QLIKE improvements over HAR are modest in magnitude across all classes (0.0–0.4%), but win rates are high (50–100%) and MCS inclusion rates are 100% for equities, currencies, and commodities, though lower for fixed income (50%), where Treasury yield volatility (TNX, TYX) favors the observable-transition specification at the weekly horizon. This pattern—consistent gains across most markets with identifiable exceptions—is consistent with a general regime smoothing mechanism, while acknowledging that the

VRNN's latent state inference is less effective for certain fixed income dynamics.

Third, we apply the Fissler-Ziegel joint elicitability framework for Expected Shortfall evaluation, achieving 1.5% lower FZ0 loss than HAR at the 1% VaR level while maintaining proper VaR calibration (Kupiec $p > 0.05$ for all models). While the FZ0 improvement is modest in unconditional magnitude, it is consistent across coverage levels and the full asset panel. This addresses the methodological gap whereby most deep-learning volatility studies employ invalid scoring rules for ES comparison, rendering their tail risk claims formally unfounded. As discussed in Section 5.4, unconditional risk metrics inherently dilute regime-dependent advantages by averaging over predominantly calm periods; the true economic value of latent regime identification for tail risk may be substantially larger during stress episodes.

The Deep-LSTR framework bridges deep probabilistic inference and interpretable econometrics, providing regime-dependent coefficients that can be examined and tested while leveraging latent state inference to filter noise from regime signals. Unlike pure black-box approaches, the STR-HAR structure preserves economic interpretability: practitioners can examine how volatility persistence parameters shift between regimes and validate that estimated dynamics accord with economic intuition. The regime smoothing mechanism—whereby the latent state appears to act as a de-noised regime indicator—represents a theoretically motivated interpretation of Deep-LSTR's forecasting gains, distinct from claims of anticipatory capacity.

Several limitations warrant acknowledgment. First, the linear projection from the multi-dimensional latent state to a scalar regime indicator ($s_t = \alpha^\top z_t$) constrains the regime to be a one-dimensional summary of the latent space. This is a deliberate design choice: it

ensures compatibility with the STR-HAR framework and preserves interpretability of the transition function. However, it limits the regime's capacity to capture multi-dimensional latent structure. We emphasize that the VRNN learns z_t freely via the ELBO objective—the projection is applied after inference, not during training—so the latent representation is not constrained to mimic any observable. The empirical evidence supports this distinction: the latent indicator normalizes 2 to 11 days faster than the observable proxy during crisis episodes, a property that a simple smoothed volatility measure cannot produce. Future work could explore nonlinear projection or end-to-end training that jointly optimizes latent inference and forecasting. Second, the deep state-space model requires more computational resources than standard HAR estimation. VRNN training requires approximately 10-20 minutes per asset on Apple M4 Pro hardware for the full training schedule, though early stopping typically terminates within 200-300 epochs. However, the annual retraining protocol limits this burden to once per year, after which daily forecasting requires only forward inference through the trained network (under 1 second per asset). For a 21-asset portfolio, annual model updating completes within 4-6 hours—comparable to sophisticated GARCH variants with numerical MLE optimization—and daily production forecasting adds negligible latency. Third, the two-dimensional latent space, while optimal in our robustness analysis, may not capture all relevant regime dynamics in more complex market environments. Fourth, our analysis focuses on univariate volatility forecasting; extending to multivariate settings with cross-asset spillovers remains for future work.

Future research directions include incorporating additional information sources such as options-implied volatility surfaces and high-frequency order flow data into the latent state

inference. Alternative deep state-space architectures, including attention-based transformers and neural ordinary differential equations, may further improve regime identification. Finally, extending the framework to joint multivariate volatility and correlation forecasting would enhance portfolio risk management applications.

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Appendix

A. VRNN Architecture Details

Strict Causality of Latent Inference: Latent inference is performed with a forward (filtering) GRU; the latent state at time t depends only on inputs up to t. Computing the sequence in one forward pass does not introduce future information. Specifically, the approximate posterior $q(z_t|x_{\leq t}, z_{<t})$ conditions only on observations through time t, maintaining

the causal structure required for valid out-of-sample forecasting.

Table A1: Neural Network Architecture

Component	Layer	Dimensions	Activation
GRU Encoder	Recurrent	$3 \to 16$ (hidden)	GRU gates
	Output (μ)	$16 \to 2$	Linear
	Output ($\log \sigma$)	$16 \to 2$	Linear (clamped to $[-8, 4]$)
MLP Decoder	Input	$2 \to 32$	Tanh
	Hidden	$32 \to 32$	Tanh
	Output ($\hat{x}$)	$32 \to 3$	Linear
Prior	Diagonal AR(1)	$z_t = \rho \odot z_{t-1} + \eta_t$	$\rho = \tanh(\rho_{\text{raw}})$
	Innovation σ_η	2 (learnable)	$\exp(\cdot)$
Observation noise σ_x		3 (learnable)	$\exp(\cdot)$

Notes: The GRU encoder directly produces approximate posterior parameters (μ, σ) for the latent state $z_t \in \mathbb{R}^2$. The prior is a diagonal AR(1) process with learnable persistence $\rho \in (-1, 1)$ and innovation variance, initialized at $\rho_{\text{raw}} = 0.3$. The decoder reconstructs observations from the latent state without conditioning on the GRU hidden state. No dropout is applied.

B. Optimization Details

Training Configuration: - Optimizer: Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$) - Learning rate: 2×10^{-3} with weight decay 10^{-4} - Gradient clipping: max norm = 5.0 - Training mode: full-sequence (single pass over entire training window per epoch) - Epochs: 600 (maximum) - Early stopping: patience = 60 epochs

STR-HAR Estimation: - Method: Basin-hopping global optimization - Local optimizer: L-BFGS-B - Basin-hopping iterations: 5 - Temperature: 1.0 - Bounds: [0.01, 12], [-5, 5]

Parameter Bound Justification: The smoothness parameter is constrained to [0.01, 12] following recommendations in the STR literature (Teräsvirta 1994; Dijk et al. 2002). The lower bound = 0.01 prevents the transition function from degenerating to a constant (eliminating regime-switching behavior), while the upper bound = 12 prevents excessively abrupt transitions that would render the model numerically unstable and approach a threshold specification where gradient-based optimization fails. The coefficient bounds [-5, 5] prevent parameter explosion during optimization while accommodating the typical range of HAR coefficients observed in realized volatility applications; preliminary estimation on our sample confirmed all converged coefficients fall well within these bounds.

Hyperparameter Selection:

The VRNN architecture employs a deliberately parsimonious design to prevent overfitting given the relatively short training windows (~600 observations per asset). The GRU hidden dimension (16 units) and decoder hidden dimension (32 units) were selected via prelimi-

nary grid search over {8, 16, 32, 64} on held-out validation loss, with smaller architectures preferred when validation performance was comparable. The learning rate (2 x 10^-3) and weight decay (10^-4) follow standard practices for Adam optimization in variational models. The KL divergence term in the ELBO and the AR(1) prior jointly regularize the latent dynamics, encouraging smooth temporal evolution without requiring explicit dropout. The two-dimensional latent space (d=2) was validated in Section 6.1, with higher dimensions showing degraded out-of-sample performance consistent with overfitting. This parsimonious architecture ensures the model generalizes across the diverse assets in our panel while remaining computationally tractable for annual retraining.

Reproducibility: - Random seed: 123 (fixed for all stochastic operations including NumPy and PyTorch) - Hardware: Apple M4 Pro (24GB unified memory) - Software: PyTorch 2.6, Python 3.12 - Training time: 10-20 minutes per asset (full 600 epochs); early stopping typically terminates within 200-300 epochs

Code and Data Availability: Replication code is publicly available at https://github.com/yenwee/latent-regime-thesis. The repository includes deterministic scripts to download all public data used in the analysis, a locked software environment (`requirements-lock.txt`) with pinned package versions for exact reproducibility, and configuration files reproducing all reported results.

CUDA Determinism Settings: All experiments were conducted on CPU, which provides deterministic behavior given fixed random seeds. For users seeking to replicate results on CUDA-enabled GPUs, the following determinism settings should be enabled:

```
import torch
```

```python
import os

# Set CUDA determinism environment variable
os.environ["CUBLAS_WORKSPACE_CONFIG"] = ":4096:8"

# Enable deterministic algorithms globally
torch.use_deterministic_algorithms(True)

# Ensure cuDNN determinism
torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False
```

These settings ensure reproducible results across runs at the cost of potential performance degradation. The `CUBLAS_WORKSPACE_CONFIG` environment variable must be set before any CUDA operations. Note that some PyTorch operations lack deterministic implementations on CUDA; if encountered, users may need to replace these operations with deterministic alternatives or accept minor numerical variations. For exact replication of reported results, we recommend using CPU execution as in the original experiments.

C. Additional Results Tables

Table A2: Individual Asset QLIKE Results (H=5)

Asset	Class	HAR	STR-OBS	Deep-LSTR	GARCH-t	EGARCH-t	Best
Dow Jones	Equity	-9.091	-9.091	**-9.093**	-8.855	-8.856	Deep-LSTR
S&P 500	Equity	-9.111	-9.111	**-9.115**	-8.829	-8.829	Deep-LSTR
NASDAQ-100	Equity	-8.504	-8.504	**-8.506**	-8.230	-8.238	Deep-LSTR
Russell 2000	Equity	-8.446	-8.447	**-8.448**	-8.150	-8.191	Deep-LSTR
Energy	Sector	-7.818	-7.818	**-7.820**	-7.650	-7.650	Deep-LSTR
Financials	Sector	-8.485	-8.486	**-8.488**	-8.270	-8.281	Deep-LSTR
Technology	Sector	-8.304	-8.305	**-8.306**	-8.060	-8.072	Deep-LSTR
Utilities	Sector	-8.429	-8.430	**-8.430**	-8.332	-8.325	Deep-LSTR
7-10Y Bond	Fixed Inc.	-11.209	-11.210	**-11.211**	-10.680	-10.741	Deep-LSTR
20+Y Bond	Fixed Inc.	-9.472	-9.472	**-9.473**	-9.026	-9.083	Deep-LSTR

Asset	Class	HAR	STR-OBS	Deep-LSTR	GARCH-t	EGARCH-t	Best
10Y Treasury	Fixed Inc.	-7.443	**-7.539**	-7.524	-7.186	-7.248	STR-OBS
30Y Treasury	Fixed Inc.	-7.838	**-7.931**	-7.910	-7.641	-7.696	STR-OBS
AUD/USD	Currency	-9.199	-9.206	**-9.206**	-9.052	-8.979	Deep-LSTR
EUR/USD	Currency	-9.884	-9.886	**-9.887**	-9.787	-9.784	Deep-LSTR
GBP/USD	Currency	-9.540	**-9.541**	-9.541	-9.459	-9.447	STR-OBS
USD/JPY	Currency	-9.669	-9.672	**-9.674**	-9.544	-9.562	Deep-LSTR
Crude Oil	Commodity	-6.789	-6.790	**-6.791**	-6.682	-6.675	Deep-LSTR
Gold	Commodity	-8.957	**-8.992**	-8.820	-8.909	-8.898	STR-OBS
Copper	Commodity	-8.322	-8.485	**-8.496**	-8.093	-8.083	Deep-LSTR

Asset	Class	HAR	STR-OBS	Deep-LSTR	GARCH-t	EGARCH-t	Best
Natural Gas	Commodity	-5.970	-5.971	**-5.972**	-5.845	-5.857	Deep-LSTR

Notes: QLIKE loss (x100). Bold indicates best model for each asset.

D. Two-Stage vs. Joint Estimation

We deliberately separate latent state inference from volatility forecasting rather than jointly optimizing the VRNN and STR-HAR parameters end-to-end. Joint estimation risks *posterior collapse*, a well-documented phenomenon whereby the latent variable degenerates to the prior and the decoder learns to ignore it entirely (He et al. 2019; Wang et al. 2021). When paired with powerful decoders that can model the target directly, VAE-style models often converge to degenerate local optima where the approximate posterior equals the prior and the latent representation carries no information.

In our setting, a powerful STR-HAR decoder optimizing forecast loss would incentivize the latent state to collapse into a nonlinear proxy for lagged realized volatility, losing its capacity to capture regime dynamics distinct from observable signals. Preliminary experiments confirmed this concern: joint training produced latent states with correlation exceeding 0.95 to contemporaneous log-RV, effectively eliminating the leading indicator property documented in Section 5.5. The latent state became redundant with observable inputs rather than capturing anticipatory regime information.

The two-stage approach mirrors factor-augmented forecasting frameworks (Stock and Watson 2002), where latent factors are first extracted via principal components or state-space methods, then used as regressors in forecasting equations. This separation ensures the transition variable captures regime information through the generative model's reconstruction objective—which encourages learning market dynamics—rather than directly optimizing forecast loss, which would encourage mimicking lagged volatility.

E. Data and Code Availability

Daily OHLC price data were obtained from the Yahoo Finance API (https://finance.yahoo.com/). The dataset covers 21 liquid financial assets across equity indices, equity sectors, fixed income, foreign exchange, and commodity markets from January 2015 through December 2025.

Replication code implementing the Deep-LSTR framework, including VRNN training, STR-HAR estimation, and all evaluation procedures, is publicly available at https://github.com/yenwee/latent-regime-thesis. The repository includes: - Python implementation of the VRNN architecture - STR-HAR estimation routines - Evaluation scripts for QLIKE, FZ0 loss, and Model Confidence Set procedures - Configuration files reproducing all reported results - Deterministic scripts to download all public data - Locked software environment for reproducibility

Raw price data can be obtained directly from Yahoo Finance using the provided asset tickers.